Machine Learning for Imaging – Coursework Report

Mammographic density prediction

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1 Part A: Choose a sensible data augmentation pipeline

Our preliminary analysis of the RSNA-SMBC dataset revealed a significant class imbalance: categories B and C are over-represented compared to categories A and D (see Figure 1). We noted that the provided codebase effectively addresses this via an `ImbalancedDatasetSampler` within the training dataloader, ensuring balanced mini-batches during optimization.

To further improve generalization, we implemented a custom augmentation pipeline using `torchvision.transforms.v2`, integrating several geometric and photometric transformations:

- Geometric Transformations: we prioritized horizontal flipping, as it reflects the anatomical symmetry between left and right breasts. We also included RandomAffine (rotation of $\pm 15°$, scaling, and translation) and RandomResizedCrop to simulate variations in patient positioning and clinical acquisition.
- Photometric Transformations: we introduced stochastic adjustments to contrast, sharpness, and blur. These simulate the technical diversity inherent in different mammography hardware and exposure settings.

By combining these transforms, we encourage the model to focus on robust morphological features, thereby reducing overfitting to the training distribution.

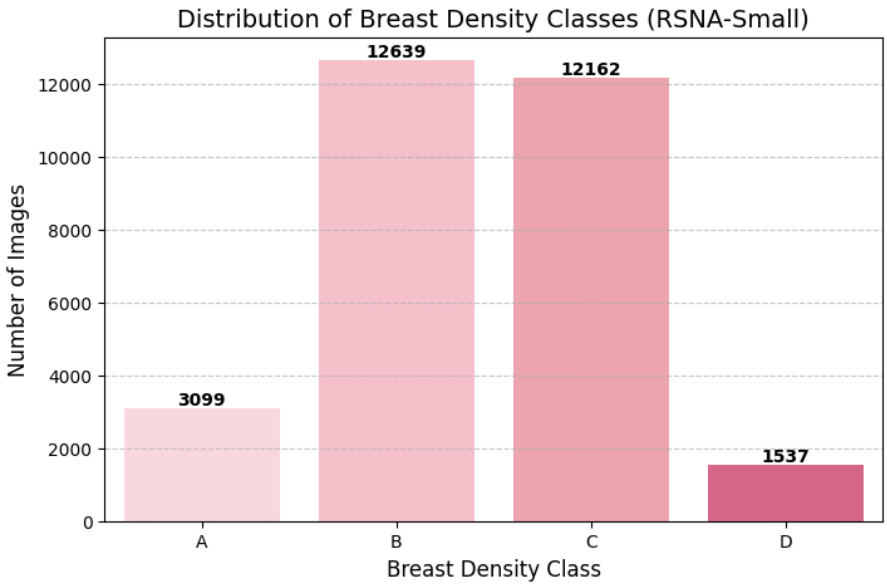


Figure 1: Distribution of breast density classes (RSNA-Small)

2 Part B: Implement a sensible model for image classification

1. Model Choice

To address breast density prediction problem, we evaluated three distinct architectures implemented via PyTorch Lightning:

Baseline MLP: a two-layer Multi-Layer Perceptron (18.9M parameters) was used as a performance baseline.

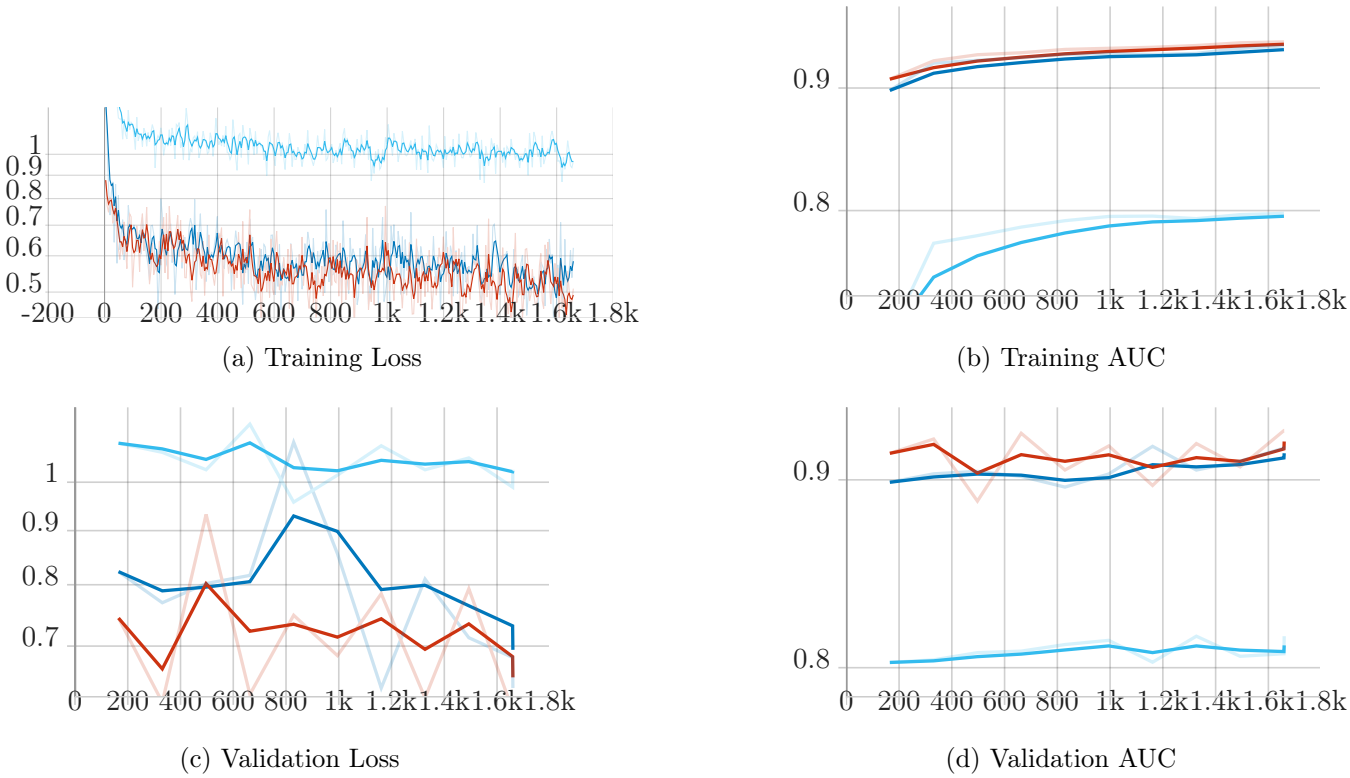
ResNet-18: a residual network (11.2M parameters) chosen for its efficiency and ability to mitigate vanishing gradients through skip connections.

DenseNet-121: a dense architecture (7M parameters) that maximizes feature reuse by connecting each layer to every subsequent layer, offering a higher parameter efficiency.

For both CNNs, we used transfer learning with "IMAGENET1K_V1" weights. This allows the models to leverage pre-trained low-level features (edges, textures), accelerating convergence and improving performance on our medical imaging task even though it is not the same style fo images.

2. Training & Validation

We trained each model for 10 epochs. Figure 2 illustrates the evolution of loss and AUC metrics during training and validation phases, while Table 1 summarizes the peak performance for each architecture.



(a) Training Loss (b) Training AUC

(c) Validation Loss (d) Validation AUC

Figure 2: Training and Validation metrics for 3 models: 2-layer MLP (Light blue), ResNET-18 (Dark blue), DenseNet21 (Red)

Regarding the convergence, both convolutional models (ResNet-18 and DenseNet-121) significantly outperformed the MLP baseline. Notably, DenseNet-121 achieved its best validation AUC (0.9282) and lowest loss (0.6090). This suggests that the features learned via transfer learning were immediately effective for our dataset.

About the stability, the training loss for CNNs showed typical stochastic oscillations while the validation AUC remained consistently high, indicating robust generalization.

Finally, despite having the fewest parameters (7M), DenseNet-121 proved to be the most performant model, confirming the benefits of dense feature reuse for medical imaging tasks.

Model	**# of Params**	**Best Epoch**	**Validation AUC ↑**	**Validation Loss ↓**
2MLPmodel	18.9M	7	0.8160	1.0275
ResNet-18	11.2M	6	0.9191	0.6389
DenseNet-121	7M	9	**0.9282**	**0.6090**

Table 1: Performances of different models during Training and Validation

3. Test and Evaluation

Results (Table 2) align with validation metrics, showing no significant overfitting. DenseNet-121 achieved the highest AUC of 0.9201, confirming that dense architectures are highly effective for breast density classification tasks.

Model	**Test AUC**	**Test Loss**
2MLPmodel	0.7946	1.0622
ResNet-18	0.9079	0.6720
DenseNet-121	**0.9201**	**0.6186**

Table 2: Performances of different models during testing

3 Part C: Conduct a subgroup performance analysis

Per-class ROC analysis shows strong overall performance, with AUC values ranging from 0.86 to 0.97 (figure 3). Classes 0 and 3 exhibit excellent separability (AUC $\geq$ 0.95), while classes 1 and 2 show slightly lower performance (AUC $\leq$ 0.90), suggesting greater visual similarity and reduced separability.

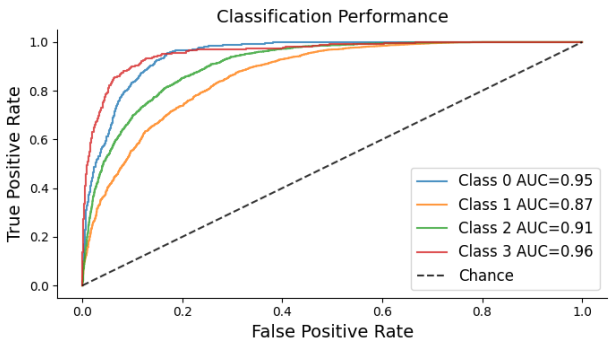


Figure 3: ROC Curves and Per-class AUC metrics

Subgroup analysis using micro-AUC (figure 4) reveals minimal differences in laterality and imaging view, indicating robust generalization across acquisition conditions. In contrast, performance decreases for implant-positive samples (AUC $\approx$ 0.89 vs 0.93 overall), likely due to imaging artifacts and altered tissue appearance.

Age-related performance differences show a clear trend. The lowest performance is observed in the youngest subgroup (40 years, AUC 0.886), which also exhibits the highest variability in the bootstrap analysis. This is likely due to the smaller sample size (n=82), which reduces the statistical reliability of the estimate. In contrast, performance improves steadily with age, reaching the highest values in the 70–79 and 80 subgroups (AUC 0.95–0.96). This trend may reflect improved class separability in older patients, potentially due to more pronounced structural differences or reduced ambiguity in imaging patterns.

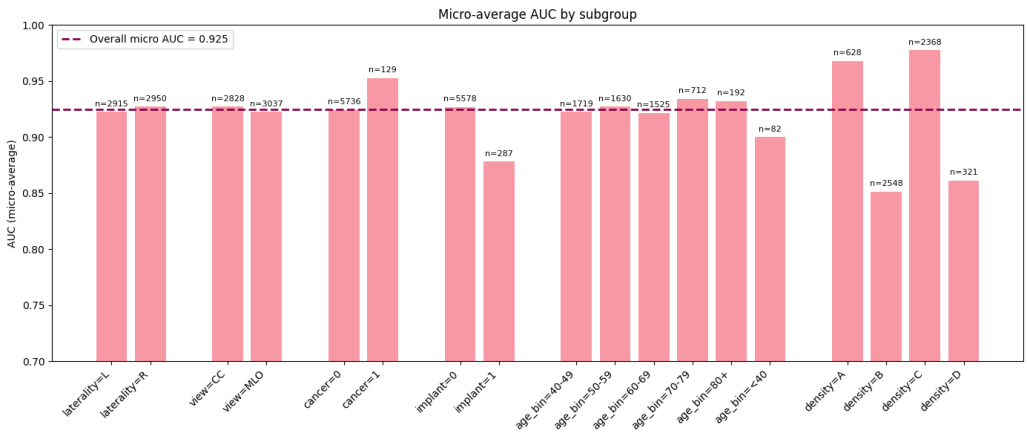


Figure 4: Micro-average AUC by subgroup

Attribute	Group	n	Micro-AUC	Δ vs overall
Overall	–	5865	0.926	0.000
Implant	1	287	0.890	-0.036
Age bin	≤40	82	0.886	-0.040
Age bin	60–69	1516	0.944	+0.018
Age bin	70–79	832	0.953	+0.027
Age bin	≥80	192	0.957	+0.031
Density	B	2548	0.893	-0.033
Density	A	628	**0.966**	+0.040
Cancer	1	129	0.959	+0.033

Table 3: Micro-AUC performance across selected subgroups compared to overall performance

Figure 5 shows the micro-AUC distributions after bootstrap resampling (B=300). It confirms that smaller subgroups exhibit higher variability, reflecting reduced statistical power.

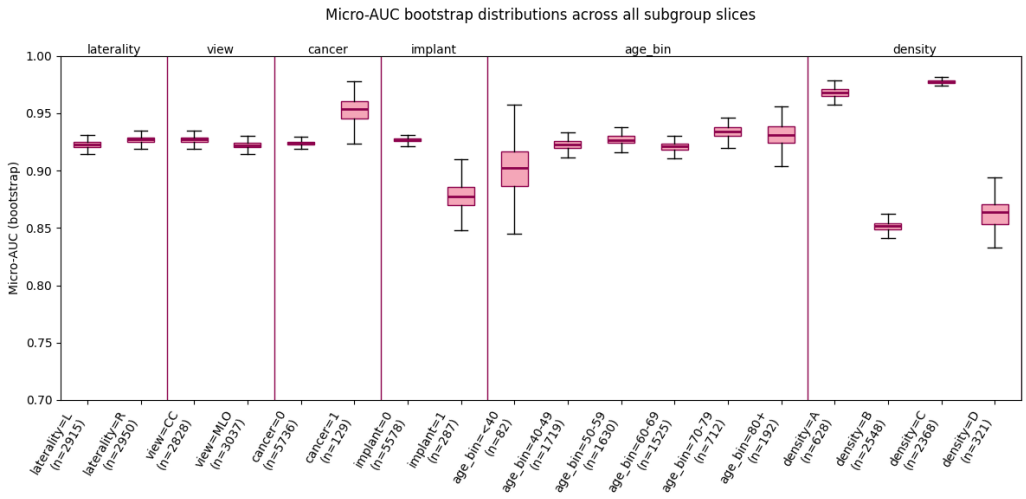


Figure 5: Micro-AUC bootstrap distributions across all sub-groups

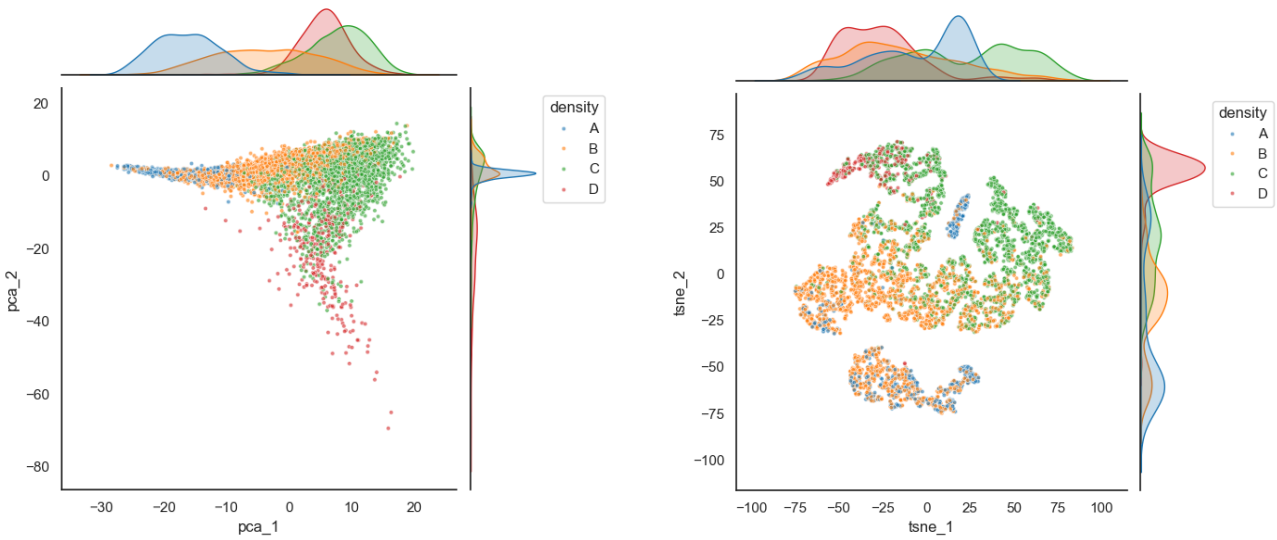
Importantly, micro-AUC is influenced by the class distribution within subgroups. Therefore, observed performance differences may partially reflect the composition of the subgroup rather than the intrinsic limitations of the model.

4 Part D: Model inspection

Model inspection was performed by projecting the 1024-dimensional feature embeddings from the DenseNet-121 model. PCA analysis revealed that the first four components capture approximately 90% of the total variance, indicating a highly efficient feature representation.

As shown in Figure 6, the PCA projection (with the first two components) reveals a continuous gradient of density, where class A (fatty) and class D (dense) occupy opposite poles. This confirms that the network has learned to represent breast density as a visual spectrum. The t-SNE visualization further demonstrates that while class D forms a distinct, highly-separable cluster, classes B and C exhibit significant overlap in the latent space. This observation provides a structural explanation for the lower per-class AUC scores (AUC ≤ 0.90) observed for these middle categories, as their morphological features are less distinct.

Interestingly, class A is distributed across several discrete clusters in the t-SNE projection, likely reflecting variations in breast anatomy or imaging acquisition that do not interfere with the model's ability to identify fatty tissue.



(a) PCA Projection

(b) t-SNE Projection

Figure 6: Dimensionality reduction of feature embeddings colored by breast density (A-D). The marginal distributions highlight the separability of extreme classes versus the overlap of middle classes.